

Article

Beyond Sociodemographics: Attitudinal and Personality Predictors of Lexical Change

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Abstract

Moving beyond traditional sociodemographic models, this study investigates the psychometric drivers of lexical change. Using Swiss German as a case study, we compare historical data from the *Sprachatlas der deutschen Schweiz* (1939–1958) with a recent large-scale app-based survey (N = 1013) to quantify trajectories over the past century. We identify four distinct mechanisms: exogenous convergence (*Schmetterling*), endo-normative leveling (*Rande*), endogenous innovation and divergence (*schlittschuhlaufen*), and diachronic persistence (*Stäge*). For the locally rooted speakers in our dataset, structural analysis indicates that traditional variables carry less weight than expected. While age remains the primary vertical predictor, psychological factors outperform traditional variables (e.g., gender, social networks) in this environment of ubiquitous exposure. Multivariate models demonstrate that lexical choices are strongly influenced by individual disposition: traits such as agreeableness accelerate the adoption of supraregional forms, whereas a strong local identity functions as a “brake” against standardization. Ultimately, while macro-factors create the pressure for change, individual micro-factors determine whether it takes hold. A speaker’s attitude acts as a “filter” and their personality as a “gate,” deciding whether they accept or resist new forms. These findings challenge purely structural accounts, suggesting that for these locally rooted speakers, even without high physical mobility, lexical change is shaped by a psychometric architecture.

Keywords: Swiss German; lexical change; dialect leveling; language attitudes; personality traits; individual differences



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1. Introduction

Lexical variation provides a clear window onto processes of language change and the social dynamics that underlie them. Historically, variationist research has concentrated heavily on phonology, as phonetic variables are highly frequent and largely subconscious, making them ideal for modeling gradual structural change (cf. [Kerswill, 2010](#)). Consequently, the lexicon has attracted less sustained attention, with many sociolinguists keeping lexical data “at arm’s length” due to its typically conscious nature and susceptibility to rapid borrowing. However, we argue that this exact characteristic—the conscious, salient nature of word choice—makes lexical variation the ideal testing ground for investigating individual psychological drivers. Because lexical choices are closely tied to identity and language attitudes, they offer a direct window into how personality and attitudinal factors shape linguistic behavior.

This relative neglect of the lexicon is especially striking in settings where this exact dynamic is highly visible, such as in German-speaking Switzerland. This sociolinguistic situation—referred to as ‘medial diglossia’—is defined by a functional split between two language varieties (cf. [Ferguson, 1959](#); [Rash, 1998](#)). Swiss German dialects serve as the everyday spoken language across all social strata and education levels. In contrast, Standard German functions mostly as the written language and is spoken primarily in formal settings, such as schools and media broadcasts (although dialect is becoming the new norm among younger people in texting, for example, cf. [Androutsopoulos and Ziegler \(2019\)](#), [Felder \(2020\)](#), and [Leemann \(n.d.\)](#)). Because dialects are the default for daily spoken interaction, this structural division has helped, to some degree, maintain regional vocabulary even amid increasing mobility and urbanization. This study investigates the interplay of lexical choice, identity, and usage, focusing on change over the past century. Drawing on historical material from the *Sprachatlas der deutschen Schweiz (SDS)* and comparing it with contemporary survey results, we trace which dialect words have persisted, which have shifted, and why. We discuss processes of convergence to the Standard, emergence of national forms, diffusion of local forms, and stability, and argue that the lexicon—often sidelined—provides an important perspective on both continuity and change. We also show that attitudinal and personality-related factors play a major role in shaping the dynamics of lexical change. In doing so, we challenge what might be called a ‘demolitionist’ view—the idea that dialect change is just a simple process of words ‘dying out’ or being ‘destroyed’ by the standard language. Instead, we argue that the changes we see are actually a complex restructuring, driven by the individual choices and personal outlooks of the speakers themselves.

Of course, these individual choices do not happen in a vacuum. Broad structural macro-factors—such as media exposure, migration, and the economic pull of urban centers like Zurich—create the overarching pressure for dialect shift. However, while these macro-factors create the pressure, it is—so we argue—the micro-factors that strongly contribute to the actual adoption. Specifically, we propose that a speaker’s individual attitude acts as a ‘filter’ for adopting standard forms, while personality acts as a ‘gate’ for accepting new regional trends.

2. Literature Review

2.1. Lexical Variation in the German-Speaking Area

Lexical variation has been at the heart of German dialectology from the very beginning. Long before sociolinguistics emerged, dialect atlases systematically charted regional vocabulary. Wenker’s 19th-century surveys laid the groundwork for the *Sprachatlas des Deutschen Reiches* (‘Linguistic Atlas of the German Empire’, [Wenker, 1889–1923](#)), which revealed remarkable lexical diversity—often dozens of words for the same concept—at the village level. In Switzerland, this tradition culminated in the *Sprachatlas der deutschen Schweiz (SDS)*, directed by Hotzenköcherle and published in eight volumes between 1962 and 1997 ([Baumgartner & Hotzenköcherle, 1962–2003](#)). Based on mid-20th-century fieldwork with older rural speakers, the *SDS* includes over 1500 maps, many lexical, and offers a fine-grained record of pre-modern Swiss German vocabulary. Together with the *Idiotikon* (‘Swiss German Dictionary’, [Idiotikon, 1881](#)), it remains a central resource for traditional dialect lexis. While the *SDS* originally provided a largely static, descriptive snapshot of the dialect landscape—meaning it captured the speech of a single older generation at one specific historical moment—in the context of the present study, it serves as our historical anchor and primary point of comparison. By using this comprehensive mid-20th-century dataset as a baseline, we are positioned to quantify the exact trajectories of lexical change over the past 70 years.

From the late 20th century onwards, variationist research began to study change in dialects over time. While phonology remained the dominant focus, a number of studies examined lexical shift. For contemporary German, Möller and Elspaß (2021), for example, examined the extent to which regional lexical diversity persists and how current distributional patterns can be explained. Using data from the *Atlas zur deutschen Alltagssprache* (AdA, ‘Atlas of German everyday language’, Elspaß & Möller, 2003)—a long-running crowd-sourced online project—Möller and Elspaß compare present-day patterns with earlier evidence from the *Wortatlas der deutschen Umgangssprachen* (WDU, ‘Word atlas of colloquial German’) collected in the 1970s. Statistical analyses such as cluster and factor analysis show that large-scale regional contrasts—most prominently the North–South divide along the *Mainlinie* (‘the Main river line’, cf. Löffler, 2003) and national boundaries—remain stable. At the same time, many formerly localized variants have receded: for instance, earlier WDU forms for ‘potatoes’ (e.g., *Potacken*, *Knollen*, *Bodenbirnen*) have largely been replaced by *Kartoffeln* in recent AdA data. The literature also documents consolidation within subnational units, with variants such as *Schulaufgabe* (‘school assignment/exam’, Bavaria) or *Probe* versus *Prüfung/Prüefig* (western vs. eastern Swiss cantons) aligning closely with administrative boundaries. They also document the diffusion of formerly localized variants, however, which often develop their own regional profiles, as seen in the South-West German use of *Hütle* for ‘traffic cone’ alongside more general forms such as *Verkehrshütchen* (‘traffic cone’). Similarly, culturally modern concepts may generate novel regional contrasts, exemplified by the national split between *Velo* in Switzerland and *Fahrrad* (‘bicycle’) in Germany and Austria.

Möller and Elspaß identify several factors driving current developments in German lexical variation. Access to printed media through current schooling, the rise in supraregional broadcast media, and the ubiquity of digital media have made supraregional, passively received linguistic models omnipresent. This facilitates *Makrosynchronisierung* (‘macro-synchronization’). Rooted in Schmidt and Herrgen’s (2011) framework of linguistic dynamics (*Sprachdynamik*), macro-synchronization describes how speakers align their language toward a supraregional norm via media and institutions. Unlike micro- or meso-synchronization, which rely on direct, face-to-face interaction within a community, macro-synchronization happens from a distance. As Möller and Elspaß (2021) note, this distant alignment—primarily toward the Standard acquired in school—has become far more influential today than in the past. As a result, the number of variants continues to decline, with small-scale forms increasingly replaced by variants with wider distribution or by standard forms. This concept of *Makrosynchronisierung* is highly relevant to the present study, as it provides a vital framework for understanding how standard convergence and lexical leveling can occur even among the locally rooted, low-mobility populations analyzed in our sample.

Looking at Austria in more detail, Lenz et al. (2023) examined lexical variation by comparing Austriacisms such as *Haube* vs. *Mütze* (both meaning ‘cap’, *Haube* being the Austriacism) and *Zündholz* vs. *Streichholz* (both meaning ‘matchstick’) across regions, generations, and registers. The results point to a steady decline of traditional Austriacisms, which are increasingly replaced by variants common in Germany, a development that can be traced across the full register spectrum from dialect to standard language. Vienna emerges as the central site of change, where Federal German variants spread most quickly.

According to Lenz et al. (2023), this change can be traced to three closely connected forces. First, Vienna’s position as the country’s political and economic center gives it disproportionate linguistic influence, allowing shifts emerging in the capital to radiate into both the Austrian standard and the surrounding Central Bavarian dialects. Second, the city’s long history of mobility and migration has produced a highly diverse population,

creating conditions in which linguistic habits mix and spread more rapidly than elsewhere. Third, external pressure from Federal German forms—circulating through media, education, and public communication—leads to the gradual replacement of Austrian variants, a process often reinforced when institutions privilege Federal German norms. Together, these factors make Vienna the key hub of ongoing change in Austrian German.

Compared to the Swiss German context, the Austrian context provides a crucial contrast. It illustrates a model where linguistic change is driven outward from a single dominant urban hub. In contrast, German-speaking Switzerland lacks one single dominating center and is shaped by a different sociolinguistic ecology. Because Swiss lexical change cannot be traced back to the influence of a single capital city, it is necessary to look beyond structural geography and investigate individual psychological and attitudinal drivers.

2.2. Lexical Change in German-Speaking Switzerland: Diffusion, Regional Leveling, and Standardization

There are a number of relatively recent and relevant studies that have looked at lexical variation and change in Swiss German. Christen (1998), for example, investigates whether young speakers converge toward neighboring dialects or toward Standard German, or whether some developments instead point toward divergence. To address this, she analyzed a corpus of informal dialogues with young speakers from across German-speaking Switzerland, complemented by data from student writing, and compared these with the historical baseline of the *Sprachatlas der deutschen Schweiz* (SDS).

Christen's findings indicate that convergence toward Standard German is most pronounced in the lexicon, where many traditional dialect words are giving way to Standard forms (e.g., *Anke* 'butter' → *Butter*). Conversely, divergence from the Standard is most evident in morphology, where processes such as apocope produce forms that are arguably more innovative than the Standard (e.g., *de Chinde* 'children [dat. pl.]' → *de Chind*). In terms of phonetics, convergence can occur not toward Standard German, but rather toward supraregional Swiss patterns. This is evident in Fribourg speakers who replace the local diphthong [ɔə] in *Brot* ('bread') with the long back lax vowel [ɔ:], a feature typical of Western Swiss German (where Federal German has [bro:t]). Consequently, phonetic and morphological features remain the stable markers of regional distinctiveness. Christen argues that these findings are closely tied to social factors: the expansion of telecommunication and broadcast media has increased dialect use and intensified contact across dialect boundaries; high mobility within a small geographic area fosters routine *polylectal* interaction; and the deformalization of public life has brought dialect into domains once reserved for Standard German (e.g., increase in dialect use in schools, cf. Werlen, 2004; Studler, 2023).

Utilizing a comparative approach between the SDS and a large-scale online survey (N > 5600), Juska-Bacher (2010) quantifies lexical change across Swiss German dialects over a 50–70-year interval. The analysis reveals that while fundamental East–West and North–South isoglosses persist, they are increasingly eroded by the diffusion of Standard German or supraregional forms. This 'invasive' trend is exemplified by the widespread displacement of traditional terms such as *Summervogel* and *Fifalter* by the Standard-aligned *Schmütterling* ('butterfly'), and by the expansion of the Standard-aligned *Zwib(e)le* ('onion') into territories historically dominated by *Böle*. Complex dynamics are further visible in the case of 'freckles,' where the Standard *Summersprosse* has overlaid the traditional East–West opposition of *Merzefläcke* versus *Laubfläcke*. Generalized Linear Mixed Effects Models identify age as the decisive sociodemographic predictor, showing a linear correlation between younger cohorts and the attrition of conservative SDS vocabulary. Statistical analysis identified parental dialect as a significant predictor for the retention of traditional forms, whereas gender and duration of residency showed no measurable effect.

More recently, [Schneider \(2022\)](#) frames lexical change in the Bernese Midlands as a complex dynamic of displacement and restructuring. In her study, too, a dominant trajectory is the substitution of traditional dialect lexemes with Standard German equivalents—evident in shifts such as *gvätterle* → *spile* ('to play') or *Moudi* → *Kater* ('tomcat')—yet this convergence is not absolute, as specific autochthonous items like *Anke* ('butter') and *Mùni* ('bull') demonstrate continued stability. Furthermore, the data suggests a counter-trend of revitalization where previously receding terms, such as *Mürggu* ('heel of bread'), are being reclaimed, likely acquiring indexical value as regional shibboleths (cf. [Eckert, 2008](#)). The lexicon is also being reshaped by mechanisms of semantic reanalysis, which facilitate the survival of older forms through functional shifts; this includes the semantic specialization of *bschütte* (narrowing from 'to water' to 'to spread manure') or the pejorative shift of *gvätterle*. Thus, lexical change in the region is characterized not merely by attrition under Standard German influence, but by active strategies of semantic functionalization and symbolic retention.

Beyond specific lexical shifts, variationist theory suggests that these changes are often stratified by traditional social categories. For instance, gender is frequently cited as a primary driver of linguistic innovation (cf. [Labov, 2001](#), [Eckert & McConnell-Ginet, 2003](#)), while higher educational attainment tends to correlate with adherence to adult-dominated institutional norms and the standard language ([Labov, 2002](#)). In the Swiss context, the high visibility of dialect in public life—and the absence of a single dominant urban center—makes it unclear whether these classic structural factors remain the primary factors of change. By testing these sociodemographic variables against individual psychological factors, we can determine whether the current wave of leveling is a broad, across-the-board phenomenon or one still restricted to specific social groups.

2.3. Language Change and Attitudinal and Personality-Related Factors

While details like a person's age or hometown help us track large-scale language changes, they do not explain why two neighbors might make very different word choices. To bridge this gap, it is crucial to look at other factors, such as individual psychological predictors. This paper uses the widely accepted 'Big Five' model of personality ([McCrae & Costa, 2003](#)), which groups behavior into five stable traits: Conscientiousness (reliability and diligence), Extraversion (sociability and assertiveness), Openness (curiosity and intellect), Agreeableness (trustworthiness and politeness), and Neuroticism (emotional stability). Past research shows that personality traits influence how people speak. For instance, in the German context, [Andresen \(2015\)](#) found that introverted speakers rely on significantly more vague expressions, whereas extraverted and introverted speakers differ qualitatively in their use of modal and discourse particles: extraverts favor particles that signal certainty, while introverts rely on those that convey hesitancy. By bringing these personality traits into dialectology, we believe we can better understand the individual choices that drive lexical change.

Beyond inherent personality traits, an individual's psychological and social orientation toward their local identity also profoundly shapes their speech patterns. For example, [Beaman et al.'s \(2021\)](#) longitudinal study of the Swabian dialect in Germany demonstrates that dialect attrition over time is actively driven or inhibited by a speaker's degree of local orientation. To quantify this, Beaman uses a "Swabian Orientation Index," which assesses a speaker's local allegiance, positive language attitudes, daily dialect usage, and even cultural competence, such as knowing how to make traditional local dishes like *Spätzle* ('traditional egg noodles'). Speakers who express a strong allegiance to the culture—proudly identifying as "real" Swabians—retain dialect features at significantly higher rates. Conversely, those who associate the dialect with negative traits, such as feeling it makes them sound "slouchy,"

are more likely to drop these regional features. Interestingly, Beaman also highlights a notable gender difference, finding that men's dialect retention is more heavily influenced by this cultural orientation, whereas women's dialect attrition is more heavily impacted by their geographic mobility. This demonstrates that regional language features are not merely geographical markers, but active tools for expressing personal identity. This leads to our two central research questions:

2.4. Research Questions

RQ1: What trajectories characterize the evolution of the Swiss German lexicon over the past century? We investigate four distinct mechanisms using representative case studies: (i) exogenous convergence toward Standard German; (ii) endo-normative leveling (the consolidation of national/Helvetic forms); (iii) endogenous innovation and divergence; and (iv) the diachronic persistence of the lexical core.

RQ2: Which factors act as determinants of this lexical stability or change? We assess the relative impact of structural sociodemographic predictors (e.g., age, gender, education) versus individual psychometric profiles (specifically, language attitudes and personality traits) in shaping a speaker's lexical choices.

3. Materials and Methods: The *Dialäktatlas* Dataset

The methodology and database utilized in this study are derived from the *Dialäktatlas* ('Dialect Atlas') project (Leemann et al., 2025) (the complete *Dialäktatlas* (as a PDF) and all maps of the atlas are available for download at: <https://dialektatlas.ch/downloads>, accessed on 12 March 2026, see Supplementary Materials). The following section details the sampling strategy for survey sites and participants, specifies the lexical phenomena selected for analysis, and outlines the project's standardized protocols for data collection, transcription, processing, and cartographic representation.

3.1. Survey Locations

To facilitate direct comparison with historical *SDS* data, the sampling frame was derived from the original *SDS* network. While a full replication of all 573 locations was not feasible due to resource constraints, a subset of 127 sites was selected via statistical sampling to ensure regional representativeness and comparability (cf. Jeszenszky et al., 2021). Larger settlements were deliberately oversampled to capture contemporary urban speech patterns. While this introduces a slight bias toward standard-convergent forms, it more accurately reflects the current sociolinguistic demographic. To accommodate administrative changes such as municipal mergers, original site definitions were in some instances expanded to include adjacent communities to ensure sufficient participant recruitment, see Figure 1.



Figure 1. The 127 survey locations as used in the present study (*Dialäktatlas*, Leemann et al., 2025, p. 15). Two-letter codes represent Swiss cantons; a full list of these abbreviations is provided at the end of the manuscript.

3.2. Informants

The sample comprised 1013 participants (N = 1013), stratified by age and gender. The younger cohort has a mean age of 26.5 years (SD = 4.9), while the older cohort has a mean age of 69.9 years (SD = 6.7). At each of the 127 sites, the study recruited four younger speakers (approx. 20–35 years) and four older speakers (60+ years), with balanced gender distribution. This design enables a three-point temporal comparison: the SDS cohort (birth years ca. 1870–1900), the older survey group (1940–1960), and the younger survey group (1985–2002). Table 1 summarizes the specific demographics and data collection periods for each of these three groups.

Table 1. Summary of the three temporal cohorts compared in this study.

Cohort Label	Data Source	Birth Years	Age at Data Collection	Data Collection Period
Historical Cohort	SDS	ca. 1870–1900	Mostly older and rural speakers	1939–1958
Older Cohort	<i>Dialäktatlas</i>	ca. 1940–1960	60+ years	2020–2023
Younger Cohort	<i>Dialäktatlas</i>	ca. 1985–2002	20–35 years	2020–2023

To control for external dialect contact, participants were required to demonstrate strong local ties—defined as long-term residence, local socialization, and at least one parent from the region—while individuals with high daily mobility (e.g., long-distance commuters) were excluded. Educational background was recorded but not controlled; consistent with demographic shifts, younger speakers proportionately held more tertiary degrees than older speakers. Recruitment strategies were adapted to local conditions: social media and digital networks were utilized in urban centers, while direct outreach via municipalities and societies was employed in rural areas. The full quota was achieved in all locations except one (Schmitten GR).

3.3. Linguistic Material

For the *Dialäktatlas* project (cf. Leemann et al., 2025), a comprehensive set of items was used to elicit phonetic, morphological, syntactic, lexical, and pragmatic data. Of the variables mapped in the atlas, 59 pertain specifically to the lexicon. Four of these are analyzed in the Section 4 as representative case studies: *Schmetterling* ('butterfly') for exogenous convergence, *Rande* ('beetroot') for endo-normative leveling, *schlittschuhlaufen* ('ice skating') for endogenous innovation, and *Stäge* ('stairs') for diachronic persistence. The general lexical item set in the atlas includes both canonical *SDS* concepts (e.g., regional terms for 'lizards' or 'dandelion') and modern neologisms reflecting cultural shifts (e.g., *Schoggistängeli* 'chocolate bar', *Föndüüchruschte* 'fondue crust'), allowing for the analysis of both archaic persistence and lexical innovation.

3.4. Data Collection

Data collection commenced in February 2020 using face-to-face interviews but transitioned to a remote protocol in March following the national COVID-19 lockdown. A custom smartphone application was developed to present textual and visual prompts. Participants recorded their responses autonomously—meaning they pressed a record button on their screen and spoke their answers directly into their smartphone's microphone—while under synchronous supervision by trained field staff via Zoom (Zoom Video Communications, Inc., 2025). For example, to elicit the term for 'butterfly' (*Schmetterling*), the app displayed a picture of a butterfly alongside the question 'What do you call this insect?' (see Figure 2). The participant then tapped the record button and spoke their local dialect term (e.g., *Summervogel* or *Pfifolter*) aloud into the device. The elicitation protocol combined picture naming, sentence completion, standard-to-dialect translation, and open-ended questions, replicating *SDS* methodology where possible to ensure maximum comparability (see Figure 2). All materials were pre-tested across age groups and regions. Following the linguistic interview, participants completed a comprehensive metadata questionnaire (30–60 min) covering demographics, mobility, and social networks, alongside psychometric and sociocultural variables—such as personality traits (Big Five, cf. De Raad, 2000), musicality, and political orientation—to enable the multivariate analysis of drivers of change. Upon full completion of both the linguistic interview and the metadata questionnaire, participants received a CHF 100 honorarium. Because all sessions were synchronously supervised, there was zero participant attrition; every participant who commenced the study completed it in full.

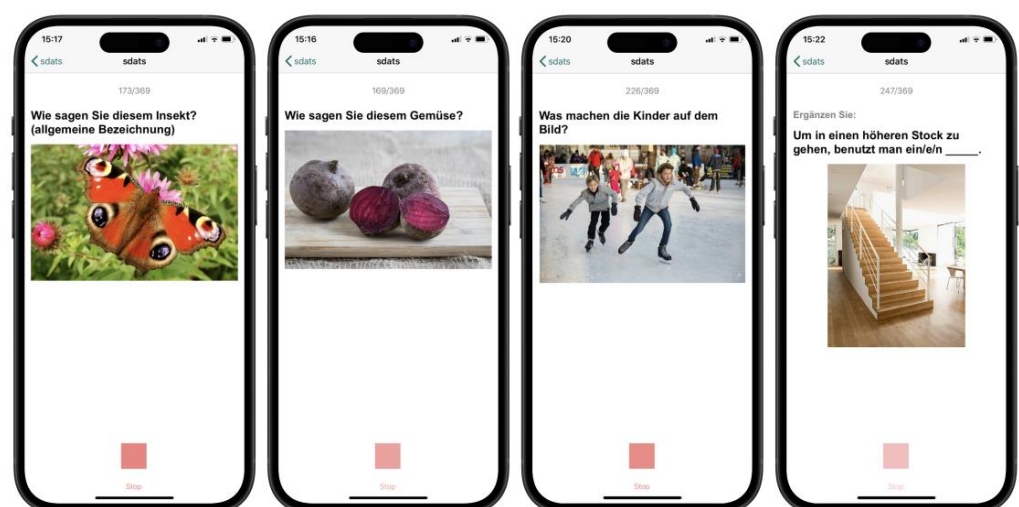


Figure 2. Prompts used for eliciting *Schmetterling* ('butterfly'), *Rande* ('beetroot'), *schlittschuhlaufen* ('ice skating'), and *Treppe* ('stairs') (*Dialäktatlas*, Leemann et al., 2025). The German text on the screens

translates to (from left to right): ‘What do you call this insect? (general term)’; ‘What do you call this vegetable?’; ‘What are the children in the picture doing?’, and ‘Complete the sentence: To go to a higher floor, one uses a _____’.

3.5. Data Processing

The recorded audio files were manually processed by trained transcribers, who listened to each response to identify the target dialect words. Phonetic transcription followed an adapted Dieth system for Swiss German (cf. Dieth, 1986). Transcriptions were phonemic and faithful to realized pronunciation, with minor orthographic adjustments for accessibility. Long vowels were indicated by doubling, vowel quality by diacritics (e.g., <è> = [ɛ] vs. <é> = [e]), and schwa represented as <e> or <a> depending on phonetic context. This system ensures a consistent and transparent representation of regional phonetic contrasts across the dataset. To facilitate cartographic analysis, lexical variants were grouped into higher-order types based on etymological and semantic similarity; for instance, the variants *Zwifaltere* (‘butterfly’), *Zwifalter*, and *Zwifälterli* were categorized under the generalized type *Zwifalter(e)*. This typological reduction balances the need for broad spatial visualization with the preservation of significant linguistic detail.

3.6. Mapping

Spatial distribution was visualized in QGIS (QGIS Development Team, 2025) using Voronoi tessellation, assigning each site a polygonal region based on geometric proximity (cf. Okabe et al., 2000). This method generates continuous dialect surfaces to indicate diffusion zones, though it necessarily abstracts away from local micro-variation. Voronoi boundaries were manually adjusted where necessary to align with known dialectal or geographic barriers. Maps display only the dominant variant(s) at each location; equal-frequency scenarios are indicated via hatching, while idiosyncratic or rare variants are omitted for readability. A standardized legend system was applied, using ‘±’ to indicate generalized forms and specific symbols for data gaps.

4. Results

4.1. Case Studies on Lexical Change

The following analysis examines four representative variables to illustrate the distinct trajectories defined in RQ1. We find exogenous convergence in the variable ‘butterfly’ (*Schmetterling*) and endo-normative leveling in ‘beetroot’ (*Rande*). These patterns are contrasted with the endogenous innovation and divergence characterizing the semantic field of ‘ice skating’ (*Schlittschuhlaufen*), and the diachronic persistence of the lexical core observed in ‘stairs’ (*Stäge*), mapping the spectrum from standardization to divergence. Beyond these representative case studies, we conclude the analysis of each mechanism by listing additional variables that demonstrate comparable trajectories. The assignment of these lexical items to the four categories is based on a systematic visual comparison of the 59 lexical maps in the *Dialäktatlas*; variables were grouped together if they exhibited spatially congruent diffusion patterns and similar rates of change relative to the historical baseline.

4.1.1. Exogenous Convergence Towards Standard German: *Schmetterling* (‘Butterfly’)

The trend of converging towards Standard German is most visibly shown in the variable ‘butterfly’. Historically, the SDS documented a highly fragmented landscape defined by a north–south isogloss: *Summervogel* (literally ‘summer bird’) predominated in the north, while *Pfifolter*-types (<OHG *fifalt(a)ra*) characterized the south. The standard-aligned variant *Schmätterling* (from East Central German *Schmetten* ‘cream’) was marginal, restricted to scattered occurrences in Bern and Eastern Switzerland—likely representing

early instances of lexical borrowing from the written standard language via schooling or early media—while micro-localisms such as *Müllervogel* (Fribourg) and *Toggeli* (Haslital) (all local terms for ‘butterfly’) further diversified the map (see Figure 3).

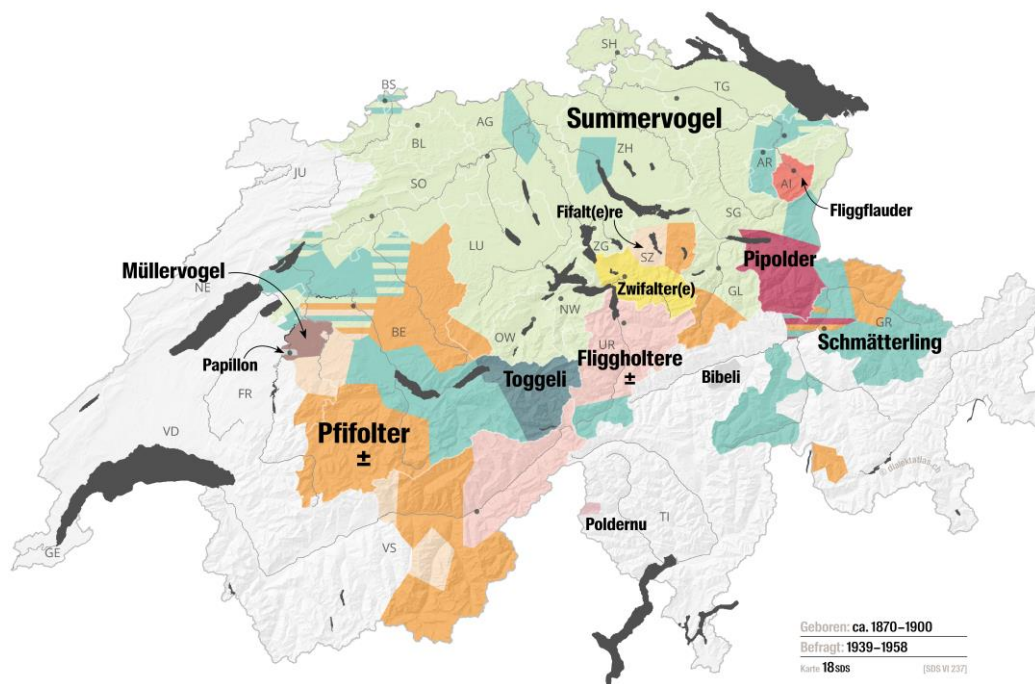


Figure 3. Distribution of variants for *butterfly* as documented in the SDS. Map legend translation: *Geboren* = Born; *Befragt* = Interviewed; *Karte* = Map.

Contemporary data shows a massive restructuring of this field. As depicted in Figures 4 and 5, *Schmättlerling* has achieved supraregional dominance. It has effectively superseded traditional lexemes and now occurs almost categorically across the entire geographic area of German-speaking Switzerland. While *Pfifolter* retains vitality in the Valais and isolated Alpine pockets, *Summervogel* has undergone significant attrition, remaining accessible only to a few speakers in Zurich as well as to some in the cantons of Aargau, Lucerne, and Obwalden (see Figures 4 and 5).

Data from the *Dialäktatlas* (Leemann et al., 2025) indicates that the standardization trajectory observed in ‘butterfly’ is systemic rather than isolated. Convergence affects a broad spectrum of semantic fields, manifesting along a continuum of replacement: in some variables, the displacement of vernacular forms is advanced, while in others the shift remains incipient. Based on the lexical variables presented in the *Dialäktatlas*, the affected lexicon includes flora and fauna (*Biene* ‘bee’, *Eidechse* ‘lizard’, *Löwenzahn* ‘dandelion’, *Heidelbeere* ‘blueberry’, *Kater* ‘tomcat’), domestic and somatic terminology (*Brille* ‘glasses’, *Schublade* ‘drawer’, *Dutt* ‘hair bun’, *Kuss* ‘kiss’, *Sommersprossen* ‘freckles’, *Schnupfen* ‘cold [med.]’, *Butter* ‘butter’, *Zwiebel* ‘onion’), as well as verbal and functional items (*giessen/begiessen* ‘to pour/water’, *flüstern* ‘to whisper’, *bekommen* ‘to receive’, *in/zu* ‘in/to [before place names]’, and *Dankeserwiderung* ‘response to thanks’). Further standard-aligned shifts are evident in items such as *Pfütze* (‘puddle’) and *Glasmurmeln* (‘marbles’). Collectively, these variables substantiate a pervasive erosion of local distinctiveness, driven by the consolidation of shared, standard-approximating norms.

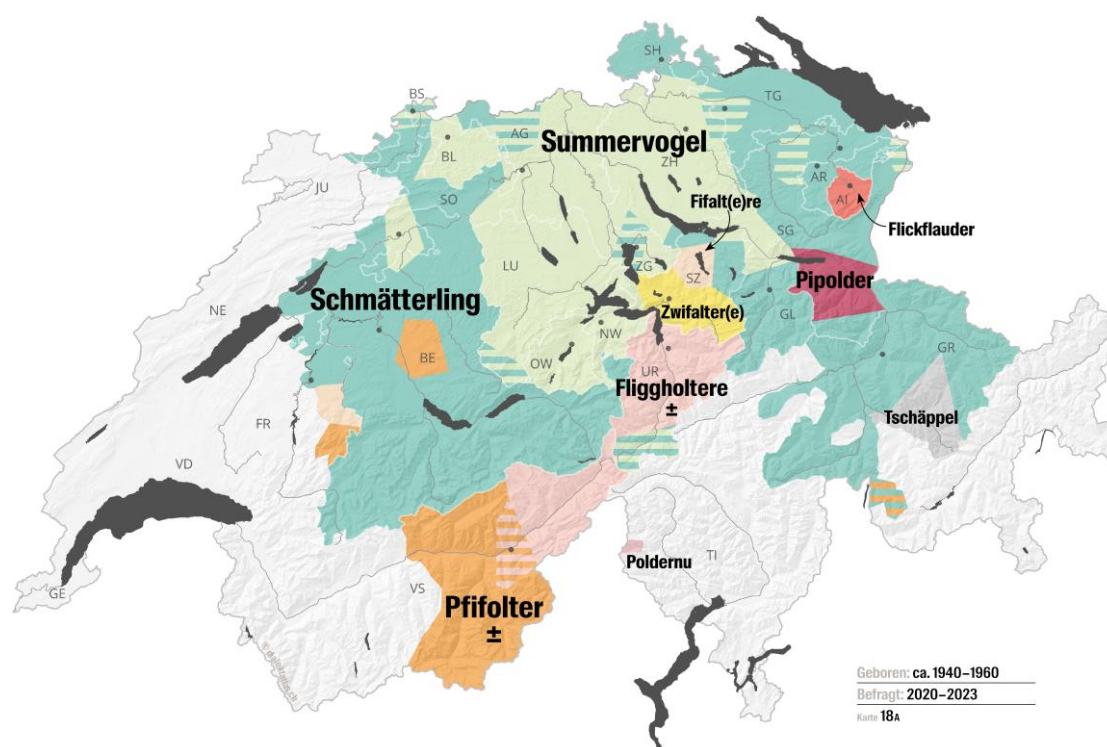


Figure 4. Distribution of variants for *butterfly* as documented in the *Dialäktatlas* older cohort. Map legend translation: *Geboren* = Born; *Befragt* = Interviewed; *Karte* = Map.

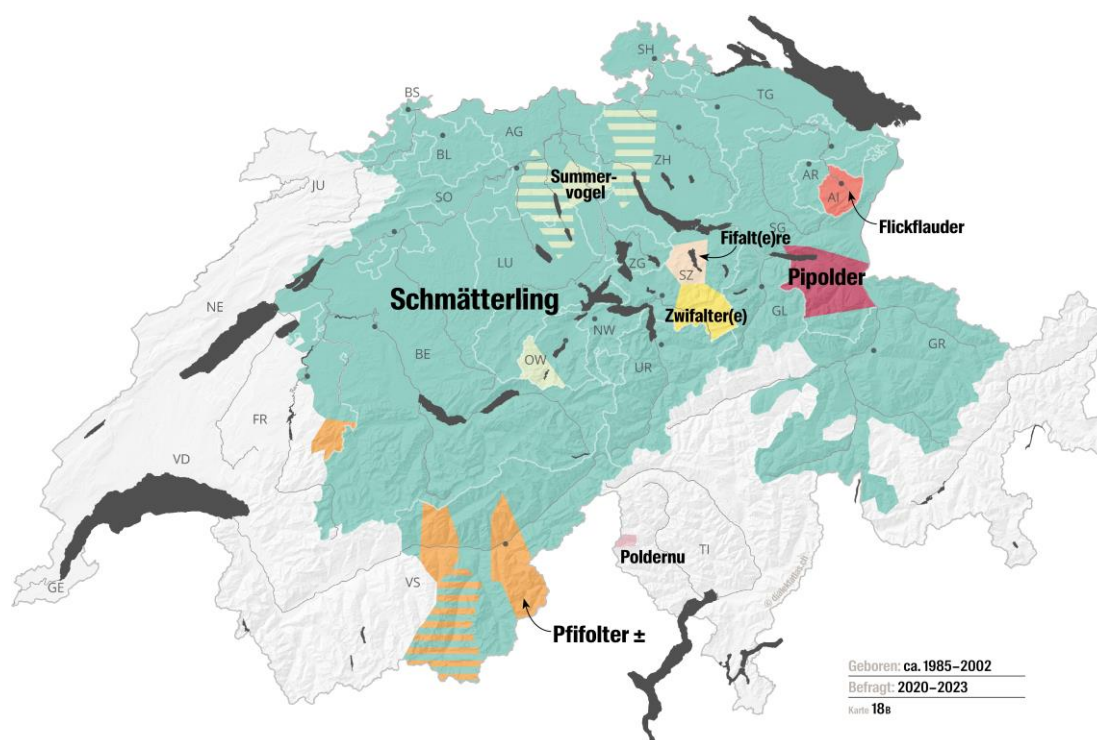


Figure 5. Distribution of variants for *butterfly* as documented in the *Dialäktatlas* younger cohort. Map legend translation: *Geboren* = Born; *Befragt* = Interviewed; *Karte* = Map.

4.1.2. Endo-Normative Leveling: Rande ('Beetroot')

A parallel evolutionary trajectory is defined by the consolidation of specifically Helvetic forms, where a single Swiss-internal variant expands to displace regional diversity. This process—distinct from Standard German convergence—is typified by the variable

‘beetroot’. Historically, the SDS recorded heterogeneity derived largely from Old High German roots (*ruoba* ‘turnip’; *retih* ‘root’; *rande*, etymologically obscure). The distribution was polycentric: *Rande* characterized the Northeast, *Raane/Roone* the Basel region, and *Räätech* the canton of Bern, while compound forms like *(roote) Räät(e)rich* and *root(i) Rüebe* dominated the Alpine regions of the Bernese Oberland, Valais, and Fribourg. The variant *Rande* was historically confined mostly to the canton of Zurich, Aargau, Central Switzerland and the Southeast (see Figure 6).

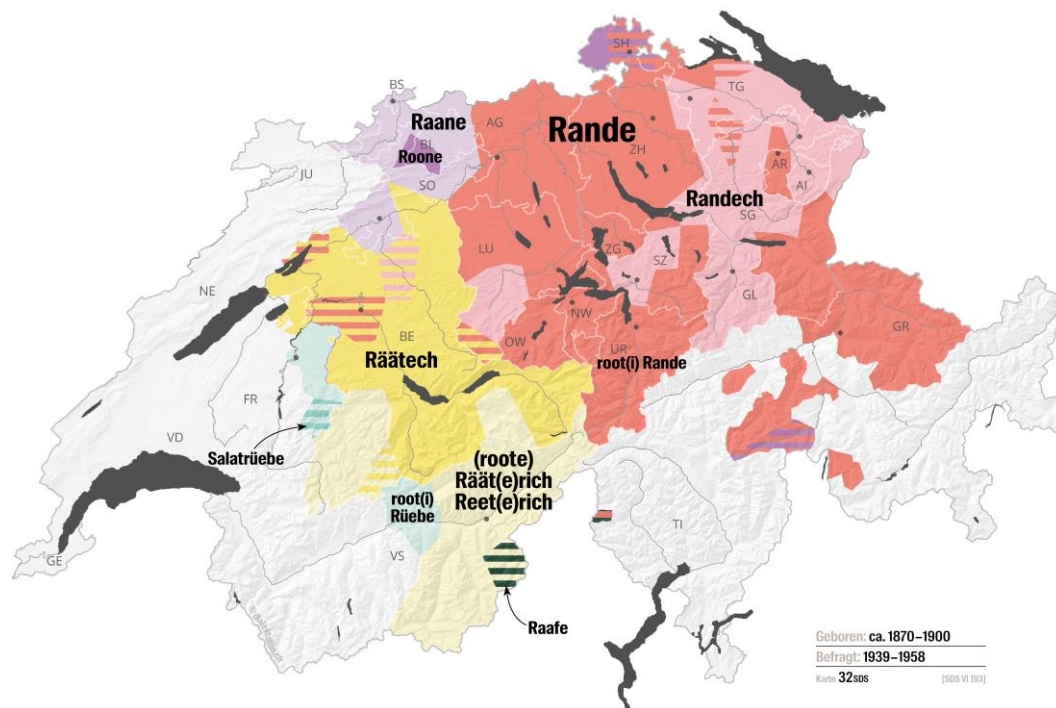


Figure 6. Distribution of variants for *Rande* as documented in the SDS. Map legend translation: *Geboren* = Born; *Befragt* = Interviewed; *Karte* = Map.

Contemporary data, however, evidences a profound leveling toward *Rande*, which has achieved near-categorical status among younger speakers across the entire dialect area (see Figures 7 and 8). While relict forms such as *Räätech* or *root(i) Rüebe* maintain some stability among older cohorts in Bern and Fribourg, localisms like *Raane* and *Rande* have been effectively erased. This shift highlights the function of *Rande* as a ‘market word’ (*Marktwort*): its prevalence in commerce and labeling has established it as the dominant supraregional dialectal norm—in opposition to the Federal German *Rote Bete* (‘red beetroot’)—facilitating its diffusion across the dialect landscape.

The *Dialäktatlas* data (Leemann et al., 2025) confirms that the trajectory observed in *Rande* is emblematic of a broader trend of endo-normative convergence, where a successful dialect variant expands to become the supraregional norm. This consolidation of specifically Helvetic forms—distinct from both the Federal German Standard and archaic localisms—is evident across multiple semantic fields. The expansion includes biological and agricultural terminology (*Hahn* ‘rooster’, *Küken* ‘chick’, *Heuschrecke* ‘grasshopper’, *Kartoffel* ‘potato’), culinary and domestic items (*Rösti* ‘hash browns’, *Käsekruste* ‘cheese crust’, *Giesskanne* ‘watering can’, *Puppe* ‘doll’, *Taschentuch* ‘handkerchief’, *urinieren* ‘to urinate’), as well as high-frequency functional vocabulary (*etwas* ‘something’). In each instance, a specific regional variant has transcended its historical isoglosses to function as a de facto Swiss variety, illustrating a mechanism of homogenization that reinforces a national dialect identity while eroding sub-national variation.

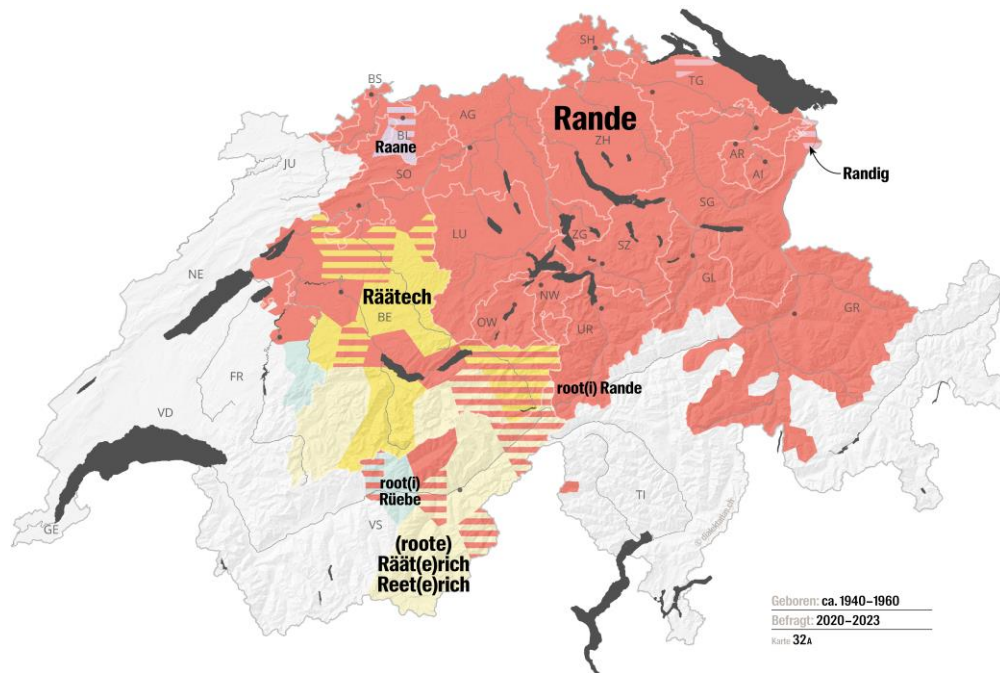


Figure 7. Distribution of variants for *Rande* as documented in the *Dialäktatlas* older cohort. Map legend translation: *Geboren* = Born; *Befragt* = Interviewed; *Karte* = Map.

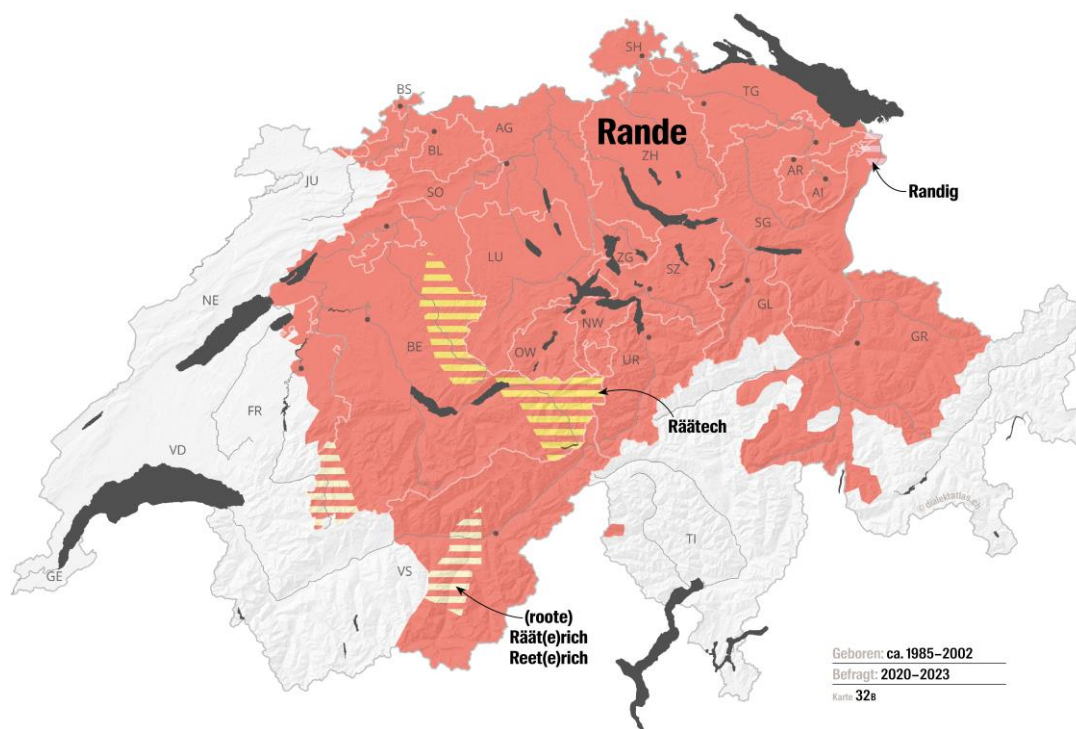


Figure 8. Distribution of variants for *Rande* as documented in the *Dialäktatlas* younger cohort. Map legend translation: *Geboren* = Born; *Befragt* = Interviewed; *Karte* = Map.

4.1.3. Endogenous Innovation and Divergence: Schlittschuhlaufen ('Ice Skating')

Beyond the trajectories of convergence and diffusion, the lexicon exhibits a distinct capacity for divergence through innovation, where novel, regionally marked forms achieve rapid spatial expansion. This dynamic is exemplified by the semantic field of 'ice skating'. Historically, the *SDS* documented a diffuse landscape characterized by morphological variability rather than sharp isoglosses. The data included periphrastic constructions

such as *Schlittschue faare* (<*Schlitten* ‘sled’) and *Schliifschue faare* (<OHG *slīfan* ‘to glide’), alongside verbal derivatives (*schliifschuene*) and unique formations (*zibis(le)*) (all local terms for ‘ice skating’). The resulting distribution was a heterogeneous patchwork lacking strong regional structuring, see Figure 9.

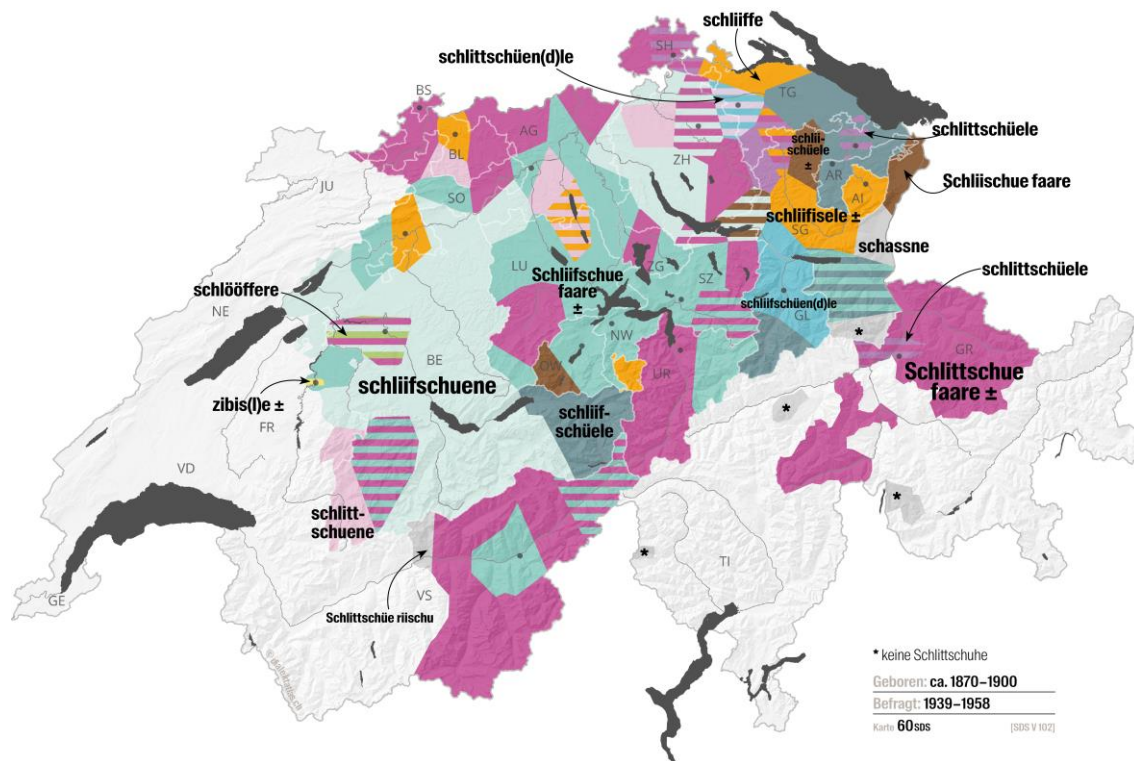


Figure 9. Distribution of variants for *schlittschuhlaufen* as documented in the SDS. Map legend translation: *Geboren* = Born; *Befragt* = Interviewed; *Karte* = Map.

Contemporary data, however, reveals a restructuring driven by morphological verbalization (see Figures 10 and 11). In Eastern Switzerland, the synthetic verb *schlittschüele* has achieved dominance, while in Central Switzerland, *schliifschüen(d)le* has expanded well beyond its historical base in the Linth Valley. The most significant innovation is observed in the Bernese region with the rapid diffusion of *schlööfle*. Likely originating as an expressive formation in youth language, *schlööfle* has effectively displaced older lexical layers to become the categorical norm among younger cohorts. This case demonstrates that lexical change is not solely a process of standardization, but can also be driven by the productivity of local dialects, creating new regional shibboleths.

Data from the *Dialäktatlas* confirm the dialect’s generative vitality, particularly through endogenous morphological innovation. This is most visible when words are shortened or given diminutive endings—a trend that often begins in youth language before stabilizing in general use. Prominent examples include the displacement of full forms by *Briüetsch* (‘brother’) and *Schwoscht* (‘sister’), alongside clippings such as *Poschi* (<*Postauto* ‘bus’), *Ufzgi* (<*Uufgabe* ‘homework’, not in *Dialäktatlas*), and *Chindsgi* (<*Chindergarte* ‘kindergarten’, not in *Dialäktatlas*). Expressive derivation drives similar innovation in affective vocabulary, visible in forms like *Bibi(i)* (‘chick’) or *Ggu(u)tchi* (‘sofa’), as well as the prolific diminutive suffixes for ‘apple core’ (e.g., *Buschi*, *Bütschgi*). Together with the distinct verbal diversity maintained in childhood play (e.g., *gireiz(le)* ‘to swing’), these variables demonstrate that Swiss German is not merely a recipient of Standard loans; instead, it actively generates socio-affectively charged forms that evolve into markers of regional identity.

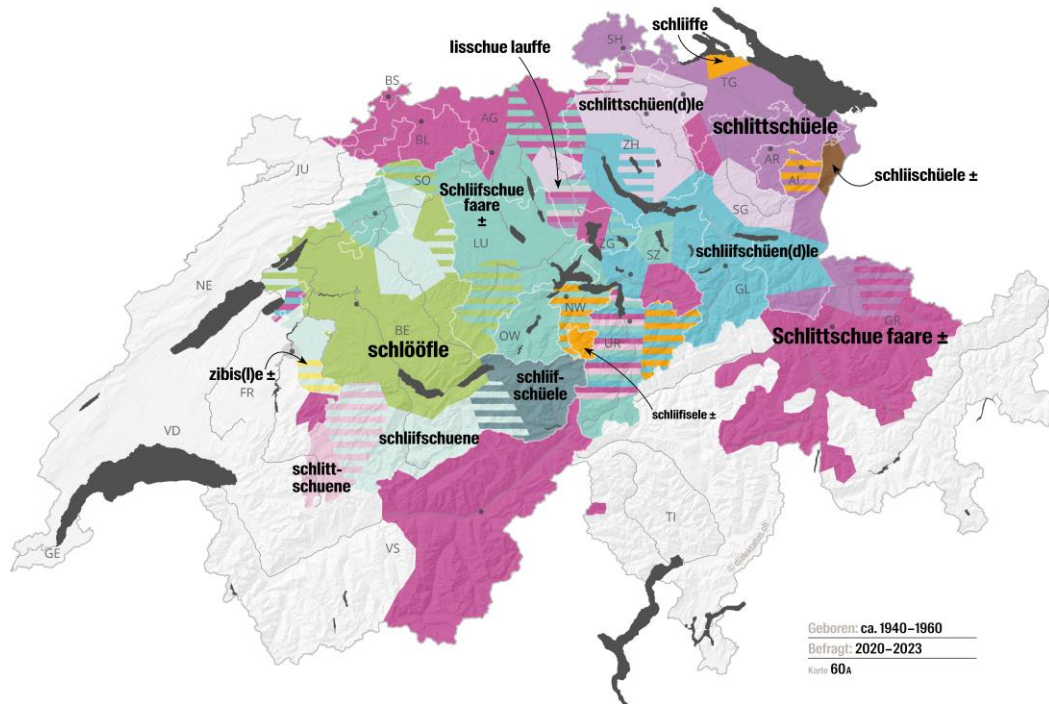


Figure 10. Distribution of variants for *schlittschuhlaufen* as documented in the *Dialektatlas* older cohort. Map legend translation: *Geboren* = Born; *Befragt* = Interviewed; *Karte* = Map.

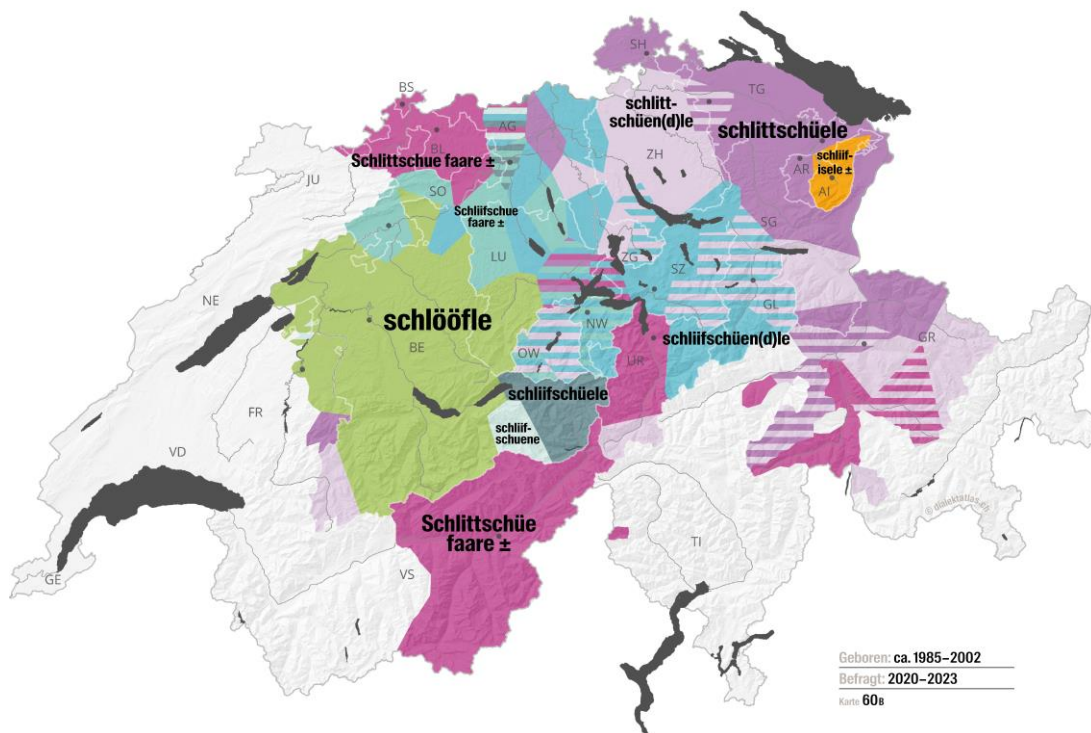


Figure 11. Distribution of variants for *schlittschuhlaufen* as documented in the *Dialektatlas* younger cohort. Map legend translation: *Geboren* = Born; *Befragt* = Interviewed; *Karte* = Map.

4.1.4. Diachronic Persistence of the Lexical Core: Stäge ('Stairs')

However, the lexicon is not uniformly susceptible to change; specific variables show remarkable diachronic stability. This resistance is epitomized by the variable ‘stairs’, where the autochthonous *Stäge* (<OHG *stēga*; cognate with *stīgan* ‘to climb’) maintains categorical dominance across the dataset. Although *Stäge* was not mapped in the original SDS—which

(*Ohrfeige* ‘slap’, *kneifen* ‘to pinch’, *Quetschfleck* ‘bruise’) alongside rural basic vocabulary (*Pferd* ‘horse’, *Zaun* ‘fence’). These variables demonstrate negligible attrition, showing that this high-frequency domestic vocabulary remains largely impervious to the leveling pressures restructuring other areas of the Swiss German lexicon.

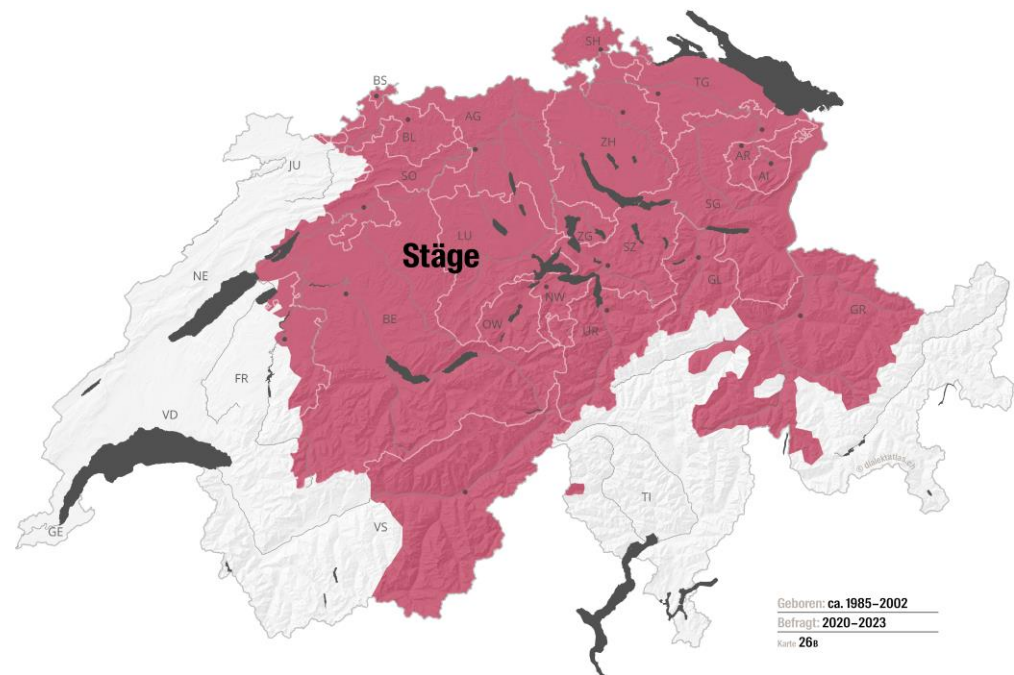


Figure 13. Distribution of variants for *Stäge* as documented in the *Dialäkatlas* younger cohort. Map legend translation: *Geboren* = Born; *Befragt* = Interviewed; *Karte* = Map.

4.2. Statistical Modelling

To quantify the determinants of lexical change, the study modeled the probability of divergence from the historical *SDS* baseline using generalized logistic regression (*Stäge*, as an illustration of diachronic stability of the lexical core, is not modeled in this section). The dependent variable was defined as a binary outcome distinguishing lexical retention (0) from change (1). The predictor space was stratified into several dimensions:

- Sociodemographic factors included exact age (*age*), education (*tertiary* vs. *non-tertiary* education), generational cohort (*age_cohort*), gender (*gender*), reference location (*reference_location*), and region (*region*; eight larger regions according to Hotzenköcherle, 1984—Bern, Southwest, Northwest, Center, Aargau, Zürich, Northeast, Grisons).
- Contact dynamics were operationalized via indices for social network integration (*SNI_PP.sqrt*, cf. Steiner et al., 2023a) and linguistic mobility (*LMI_compreh*, cf. Jeszenszky et al., 2024), capturing the density of speakers’ social ties and their history of geographic movement.
- Individual disposition was modeled through a battery of psychometric variables. Building on the theoretical framework established in Section 2.3, this included the Big Five personality traits (De Raad, 2000)—neuroticism (*neuro*), extraversion (*extra*), conscientiousness (*consc*), openness (*open*), and agreeableness (*agree*)—alongside four specific attitudinal scales derived from the questionnaire data (cf. Appendix A for detailed questionnaire):
 - *aff_identity* (local attachment, cf. Steiner et al., 2023a): A composite index measuring affective loyalty to the local variety, including items assessing pride in dialect recognition, regret regarding dialect loss, and disapproval of peer accommodation.

- *aff_pers* (interactional adaptability, cf. [Steiner et al., 2023a](#)): A measure of self-reported communicative flexibility, capturing the extent to which a speaker possesses high metalinguistic vigilance, consciously monitoring and accommodating their speech in cross-dialectal encounters versus maintaining a stable, non-adaptive dialect.
- *aff_change* (metalinguistic awareness, cf. [Steiner et al., 2023a](#)): An index reflecting the speaker’s perception of language change, including their awareness of generational differences and their evaluative concern regarding the erosion of traditional forms.
- *aff_sg* (standard orientation, cf. [Steiner et al., 2023a](#)): A scale capturing the speaker’s vertical evaluation of the dialect/standard hierarchy and their orientation toward the standard language.
- Finally, self-reported dialect practice (*DSP*, cf. [Steiner et al., 2023b](#)) was included as a behavioral covariate. Distinct from attitudinal orientation, *DSP* indexes the frequency of actual usage, aggregating self-reported language choice (dialect vs. Standard German) across different social domains (e.g., family, workplace, public administration).

Model selection employed a bidirectional stepwise fitting strategy (combining top-down and bottom-up approaches) to isolate the most parsimonious set of predictors, allowing us to determine how sociodemographic factors and individual traits jointly influence lexical change.

4.2.1. Predictors of Exogenous Convergence: Schmetterling (‘Butterfly’)

The logistic regression model analyzing the adoption of the Standard German variant *Schmetterling* revealed a robust main effect of *age_cohort*, see Table 2. Younger speakers demonstrated a significantly higher probability of convergence compared to the older generation. Crucially, however, this generational shift is significantly moderated by an interaction with *aff_identity*. This composite scale, which indexes respondents’ local attachment, pride in dialect recognition, and resistance to dialect loss, functioned as a protective factor specifically within the younger cohort. While the effect of identity was negligible for older speakers, younger informants with high *aff_identity* scores were significantly less likely to adopt the standard form. These findings suggest that while the diffusion of *Schmetterling* is structurally driven by demographic replacement, the rate of standardization is not uniform; rather, it is actively retarded among younger speakers with strong local and dialectal orientation.

Table 2. Logistic regression results for *Schmetterling* (n.s. = not significant; ** = significant).

Predictor	Estimate	Std. Error	z Value	p Value	Significance
(Intercept)	−0.214	0.547	−0.391	0.696	n.s.
<i>age_cohort</i> (younger)	2.283	0.729	3.131	0.0017	**
<i>age_cohort</i> (older) × <i>aff_identity</i>	−0.004	0.100	−0.044	0.965	n.s.
<i>age_cohort</i> (younger) × <i>aff_identity</i>	−0.251	0.091	−2.758	0.0058	**

4.2.2. Predictors of Endo-Normative Leveling: Rande (‘Beetroot’)

The logistic regression of the supralocal diffusion of *Rande* shows a distinct sociolinguistic profile driven by both demographic and psychometric factors. A significant negative main effect was observed for exact age, see Table 3, confirming that the replacement of historical variants follows a clear trajectory, with the probability of change increasing with age. Beyond this demographic baseline, two individual-level predictors proved significant. The personality trait agreeableness (*agree*) exhibited a strong positive association with lexical change, suggesting that individuals with a higher prosocial orientation or tendency toward

interpersonal accommodation are more prone to adopting the dominant “market word.” Conversely, the attitudinal scale *aff_change*—which indexes linguistic awareness of and sensitivity to dialect change—showed a significant inhibitory effect. Participants expressing higher awareness of linguistic shifts demonstrated greater retention of conservative forms. The consolidation of *Rande*, then, appears to be more than a passive demographic shift; it suggests a process modulated by individual disposition—facilitated by social agreeableness, yet potentially retarded by conscious metalinguistic awareness.

Table 3. Logistic regression results for *Rande* (n.s. = not significant, *** = highly significant, ** = significant, * = just about significant).

Predictor	Estimate	Std. Error	z Value	p Value	Significance
(Intercept)	−0.635	0.743	−0.855	0.392	n.s.
<i>age</i>	−0.015	0.0029	−5.285	1.26×10^{-7}	***
<i>agree</i>	0.602	0.204	2.954	0.0031	**
<i>aff_change</i>	−0.158	0.069	−2.294	0.0218	*

4.2.3. Predictors of Endogenous Innovation: Schlittschuhlaufen (‘Ice Skating’)

In a departure from the pervasive age effects observed elsewhere, the logistic regression for *schlittschuhlaufen* yielded no significant main effect for age cohort, see Table 4, indicating that the rate of innovation is statistically indistinguishable between generations. However, the analysis uncovered a significant interaction between age cohort and the *aff_pers* scale specifically within the older generation. Here, older speakers who self-report high interactional adaptability are significantly more likely to retain established dialect forms.

Table 4. Logistic regression results for *schlittschuhlaufen* (n.s. = not significant, *** = highly significant, ** = significant).

Predictor	Estimate	Std. Error	z Value	p Value	Significance
(Intercept)	1.200	0.272	4.416	1.0×10^{-5}	***
<i>age_cohort</i> (younger)	−0.238	0.411	−0.582	0.561	n.s.
<i>age_cohort</i> (older) $\times$ <i>aff_pers</i>	−0.233	0.083	−2.795	0.0052	**
<i>age_cohort</i> (younger) $\times$ <i>aff_pers</i>	0.103	0.085	1.207	0.227	n.s.

5. Discussion

Synthesizing the four case studies, we present a multi-path model of lexical change in Swiss German, characterized by distinct yet co-occurring trajectories

- Exogenous convergence: *Schmetterling* represents the paradigmatic shift toward Standard German, a process driven structurally by generational replacement but significantly buffered by identity orientation. Here, strong local attachment functions as a protective mechanism, retarding the adoption of standard forms among younger speakers.
- Endo-normative leveling: *Rande* illustrates the consolidation of a national Helvetic standard. This diffusion is actively facilitated by the personality trait of agreeableness, suggesting a mechanism of social accommodation, whereas a conscious awareness of linguistic change acts as a significant inhibitory factor against this supralocal spread.
- Endogenous innovation and divergence: *Schlittschuhlaufen* proves the dialect’s generative vitality through the rapid spread of the vernacular form *schlöffle*. Yet, among high-monitoring older speakers, a counter-force of “metalinguistic vigilance” persists. Rather than facilitating change, sociolinguistic competence here acts as a quality filter: these speakers actively reject perceived vernacular simplifications (“slang”) in favor of preserving the traditional, transparent compounds.

- Diachronic persistence: Finally, *Stäge* exemplifies the “recalcitrant core” of the lexicon. Despite the omnipresence of the Standard German cognate *Treppe*, the autochthonous form maintains categorical dominance across all cohorts, demonstrating that a number of words remain impervious to leveling pressures and preserve their historical integrity against broader convergence.

These findings challenge a purely demolitionist view of dialect loss, revealing that lexical change in Swiss German is not merely a function of passive demographic displacement but is actively modulated by the speaker’s psychometric architecture. The following discussion unpacks these dynamics in three dimensions. We first examine how attitudinal orientations toward identity and change serve as critical filters for standardization (Section 5.1). Subsequently, we evaluate the role of personality traits as drivers of variation (Section 5.2). Finally, we contrast these strong psychological predictors with the insignificance of traditional structural factors (Section 5.3), discussing why variables such as gender, social networks, and mobility failed to predict lexical outcomes in this dataset.

5.1. Attitudinal Determinants of Lexical Change

Addressing the objective of RQ2 to evaluate the relative predictive power of macro-social factors (e.g., age) versus individual-level attributes (e.g., attitudes), our analysis identifies attitudinal orientation as a decisive predictor of lexical outcome, secondary only to biological age. While the dominant role of age aligns with Juska-Bacher’s (2010) findings on generational replacement and Christen’s (1998) observations of lexical convergence, the current data extends these findings: individual ideological stance functions as a significant moderator of structural leveling pressures. The results demonstrate that the “social resonance” of the lexicon, alluded to in the Introduction, is empirically quantifiable—speakers’ beliefs regarding identity, language change, and adaptation actively filter the uptake of new forms.

In the case of exogenous convergence—the first mechanism outlined in RQ1-i, typified by *Schmetterling*—the *aff_identity* scale emerged as a critical brake on standardization. Among younger cohorts, strong place-attachment correlated inversely with the adoption of Standard forms. This dynamic resonates with Eckert’s (2000) observations of the ‘jock’ versus ‘burnout’ polarization, where locally oriented speakers actively resist institutional linguistic norms to construct an authentic local persona. Our results also closely align with Beaman et al.’s (2021) findings in the Swabian dialect area. Just as her ‘Swabian Orientation Index’ showed that strong cultural allegiance prevents dialect attrition, our *aff_identity* scale confirms that local pride actively protects traditional Swiss German vocabulary from standardization. This mechanism of active resistance offers a crucial corrective to structural accounts like that of Christen (1998). While Christen correctly identified the lexicon as the primary site of convergence—typically far less stable than morphology or phonology—our results demonstrate that this attrition is not a deterministic outcome. For high-identifying subjects—that is, speakers who report a strong sense of belonging and attachment to their local region—the lexicon remains a vital resource for symbolic differentiation, where specific variants are retained as deliberate, oppositional markers of belonging. The limitation of this stabilizing effect to the younger cohort suggests that even as traditional dialect socialization wanes, attitudinally grounded localism remains a potent force for retention.

Parallel dynamics appear in the leveling of *Rande* (RQ1-ii). Here, resistance to the supraregional form was predicted by *aff_change*: speakers who expressed concern about dialect loss were significantly more likely to retain traditional variants. This finding is crucial because it challenges the idea that standardization is merely a “passive reception” of new norms (cf. Möller & Elspaß, 2021). Instead, it shows that the process is negotiable: when speakers view change as a threat to their identity, their linguistic awareness acts as a social

brake. Finally, regarding endogenous innovation (*schlittschuhlaufen*), we observe a distinct ‘competence-based conservation’ effect, which refers to our measure of communicative flexibility. Surprisingly, high interactional adaptability (*aff_pers*)—typically associated with convergence—correlated here with the retention of traditional forms among older speakers. This finding requires contextualization: unlike *Schmetterling*, the innovation *schlöffle* represents a vernacular simplification rather than a prestige “standard”. In this context, we propose that for high-monitoring speakers, sociolinguistic competence manifests as stylistic propriety. Because *schlöffle* carries indexical associations with youth language or informal ‘slang,’ speakers with high metalinguistic vigilance filter it out. Thus, their adaptability manifests not as convergence, but as a defense of the ‘proper’ traditional dialect against perceived vernacular erosion.

5.2. The Role of Personality

Expanding the scope of RQ2, the analysis demonstrates that personality traits function as predictors of lexical change. This builds on existing evidence from the German context, such as [Andresen \(2015\)](#), who demonstrated how dispositional traits govern functional language use—finding, for example, that introverts rely on vague expressions to mitigate their claims, whereas extraverts project dominance through affirmative particles and the functional adaptation of discourse markers. Our results extend this logic to the domain of lexical leveling, most visibly in the *Rande* model, where agreeableness emerged as a driver of supralocalization. This finding also aligns with social-behavioral theories linking high agreeableness to conformity and interactional harmony (see [Graziano & Eisenberg, 1997](#); [DeYoung et al., 2002](#)); in the context of dialect leveling, highly agreeable speakers appear more sensitive to prevailing supraregional norms and more inclined to accommodate toward the “market word.”

The significance of these dispositional factors offers a necessary counterpoint to the macro-structural focus of earlier studies. While established research ([Christen, 1998](#); [Kerwill, 2008](#); [Lenz et al., 2023](#); [Trudgill, 1986](#)) identifies urbanization, mobility, and media as the macro-level engines of leveling, these studies typically did not include psychometric variables in their data collection. By expanding the data collection portfolio to include personality and attitudes, our results highlight the role of micro-level filters that were previously unobserved. These findings extend the nascent field of psycho-sociolinguistics (cf. [Campbell-Kibler, 2011](#); [Steiner et al., 2023a, 2023b](#)) by demonstrating that stable traits act as cognitive filters, actively modulating how speakers navigate the adoption of supraregional forms.

5.3. The Limited Impact of Contact, Gender, Education, and Macro-Regions

Because personality and attitudes proved so decisive, the lack of impact from classical sociolinguistic variables in our sample is particularly interesting. While age and region confirm that dialects retain their geographic grounding, standard factors like gender and education showed no statistical significance in predicting individual outcomes. In variationist sociolinguistics, gender, for example, is canonically posited as one of the primary vectors of change. [Labov \(2001, p. 501\)](#) famously asserted that in the majority of linguistic changes, “women are a full generation ahead of men”. Similarly, [Eckert and McConnell-Ginet \(2003\)](#) place women at the vanguard of innovation, arguing that they utilize linguistic variation to construct social distinctions. Against this theoretical backdrop, the absence of a gender effect in our models suggests that the current wave of lexical leveling in Swiss German may operate independently of gendered social practice; men and women have been adopting forms like *Schmetterling* at statistically indistinguishable rates. This lack of a gender effect contrasts not only with classic theories, but also with recent studies like [Beaman et al.](#)

(2021), who found that men's and women's dialect use is driven by distinctly different factors. Specifically, Beaman demonstrated that a man's probability of speaking dialect is heavily dictated by his local cultural orientation, whereas a woman's dialect retention is primarily impacted by her geographic mobility. Her research further concluded that as women became more geographically mobile—largely due to taking on new professional and social roles outside the home—they rapidly accommodated to standard speech, while men's dialect usage remained largely unaffected by their degree of mobility. In contrast, in our Swiss German sample, lexical leveling seems to affect both genders equally.

Interestingly, we found no effect of education: speakers with university degrees behaved no differently than those with vocational training. This contrasts with typical sociolinguistic findings, where higher education usually predicts a shift away from local dialect (Labov et al., 2013; De Decker, 2006). However, this likely reflects a selection bias in our design. By sampling only locally rooted speakers, we naturally excluded the mobile educated population who left their home communities. Yet, this limitation reveals a key insight: even among those who stay, education is no longer a prerequisite for change. The rise in forms like *Schmetterling* appears to be a broad phenomenon affecting the entire resident population regardless of academic background.

As with education, the lack of significant effects for mobility and network integration is largely an artifact of our sampling design. While our results seemingly challenge models emphasizing urban contact as the primary engine of change (e.g., Lenz et al., 2023), they are likely an artifact of range restriction inherent in our sampling design, which explicitly selected for locally rooted, low-mobility speakers. By excluding highly mobile commuters, we effectively held mobility constant, preventing it from emerging as a variable predictor. However, this limitation, too, offers a valuable insight: the observation of pervasive leveling even within this sedentary sample implies that direct physical contact is no longer the sole prerequisite for diffusion. Instead, change in these locally oriented speakers appears to be mediated by *Makrosynchronisierung* (cf. Möller & Elspaß, 2021)—a synchronization driven by digital media and national discourse rather than physical travel. This suggests that lexical change is no longer strictly stratified by traditional demographic categories, reaching even immobile speakers.

Finally, we must acknowledge a basic geographic limit: a speaker cannot adopt a new word if they have never heard it. For local innovations like *schlöffle*, someone might not use the word simply because it has not reached their town yet, rather than because they are actively resisting it. Our statistical model uses large regions, which are too broad to show exactly where a new word stops spreading. Because of this, our model cannot always tell the difference between a speaker who has never been exposed to a word and a speaker who knows the word but chooses not to use it.

5.4. Limitations and Future Directions

Certain methodological limitations warrant consideration. First, while the sampling strategy was designed to ensure comparability with the SDS, the deliberate oversampling of urban areas (compared to the historical baseline) may have captured a segment of the population more prone to convergence, potentially inflating overall rates. Second, the shift to an app-based data collection protocol during the COVID-19 pandemic introduced a potential selection bias (cf. Bethlehem, 2010); although the app was user-friendly, the format may have favored participants with higher digital literacy, a trait correlated with a higher level of education and specific psychographic attitudes (ibid.). Finally, the reliance on an apparent-time design imposes limits on interpretation. While comparing generations provides a proxy for historical change, it cannot definitively rule out age-grading—patterns specific to a life stage rather than historical change (cf. Trudgill, 2000; Sankoff, 2005). This

distinction is particularly crucial for interpreting the ‘identity brake’ observed among younger speakers. It remains an open question whether their resistance to standardization represents a permanent, lifelong orientation (true communal change) or a transient feature of adolescence and young adulthood that will fade as they enter the workforce. Confirming whether identity continues to buffer against standardization over the lifespan would require longitudinal real-time data to track these specific speakers as they age.

6. Conclusions

This study set out to investigate change in the Swiss German lexicon over the past century, questioning whether the dialect landscape is undergoing a uniform process of erosion or a more complex restructuring. By triangulating historical data from the *SDS* with a contemporary large-scale survey, the analysis reveals that lexical change is not a simple ‘demolitionist’ process of standard convergence, but a composite of four distinct processes: exogenous convergence, endo-normative leveling, endogenous innovation and divergence, and diachronic persistence. The most significant contribution of this inquiry, however, lies in its expansion of the explanatory scope of variationist sociolinguistics. We show that attitudes toward local identity can act as effective brakes on standardization, while personality traits such as agreeableness can accelerate the uptake of supraregional forms. By challenging the ‘demolitionist’ narrative, we recast the Swiss German lexicon not as a dying artifact but as a dynamic, reflexive system. Local vocabulary persists through the active, identity-driven agency of speakers who consciously navigate the tension between supralocal convergence and local distinctiveness.

Our findings suggest that while macro-factors provide the push for language change, the individual makes the final decision. Structural forces—whether urban influence, market labels, or the emotional appeal of ‘cute’ words like *schlöffle*—create the pressure to adapt. Yet, the speaker remains the ultimate checkpoint. We see that attitude acts as a filter, determining if a speaker yields to standard convergence (e.g., *Schmetterling*). Similarly, personality acts as a gate, explaining why highly agreeable people quickly adopt supraregional market words (e.g., *Rande*). Ultimately, macro-factors propose the change, but individual micro-factors decide if it is accepted.

Supplementary Materials: The complete *Dialektatlas* (as a PDF) and all maps of the atlas are available for download at: <https://dialektatlas.ch/downloads>, accessed on 27 February 2026.

Author Contributions: Conceptualization, A.L., S.K. and F.T.; methodology, F.T.; software, F.T.; formal analysis, F.T.; resources, A.L.; data curation, A.L.; writing—original draft preparation, A.L., S.K. and F.T.; writing—review and editing, A.L., S.K. and F.T.; visualization, F.T.; supervision, A.L.; project administration, A.L.; funding acquisition, A.L. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: Ethical review and approval were waived for this study. The Swiss Association of Research Ethics Committees (swissethics) confirmed on 31 January 2018, that this linguistic research project does not fall within the scope of the Swiss Human Research Act (HRA) and therefore does not require authorization by a cantonal ethics commission.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study. Prior to data collection, participants received a detailed information sheet regarding the study goals, procedures, and data protection measures. All subjects provided signed informed consent (facilitated via digital signature protocols due to pandemic-related restrictions) for their participation, the audio recording of their speech, and the transfer of data rights to the research center. Additionally,

written informed consent was obtained from the participants to publish the collected data in an anonymized form.

Data Availability Statement: The data presented in this study are openly available in the Open Science Framework (OSF) repository at <https://osf.io/vu7f8>, accessed on 27 February 2026. This repository contains the raw .csv files as well as the R scripts used for the analysis.

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Abbreviations

The following abbreviations are used in this manuscript:

Abbreviation	Meaning	English Translation/Note
AdA	<i>Atlas zur deutschen Alltagssprache</i>	Atlas of German everyday language
SDS	<i>Sprachatlas der deutschen Schweiz</i>	Linguistic Atlas of German-speaking Switzerland
WDU	<i>Wortatlas der deutschen Umgangssprache</i>	Word atlas of colloquial German
OHG	Old High German	Historical linguistic stage
LG	Low German	Historical linguistic stage
AG	Aargau	Swiss Canton
AI	Appenzell Innerrhoden	Swiss Canton
AR	Appenzell Auserrhoden	Swiss Canton
BE	Bern	Swiss Canton
BL	Basel-Landschaft	Swiss Canton
BS	Basel-Stadt	Swiss Canton
FR	Fribourg	Swiss Canton
GE	Geneva	Swiss Canton
GL	Glarus	Swiss Canton
GR	Graubünden	Swiss Canton
JU	Jura	Swiss Canton
LU	Lucerne	Swiss Canton
NE	Neuchâtel	Swiss Canton
NW	Nidwalden	Swiss Canton
OW	Obwalden	Swiss Canton
SG	St. Gallen	Swiss Canton
SH	Schaffhausen	Swiss Canton
SO	Solothurn	Swiss Canton
SZ	Schwyz	Swiss Canton
TG	Thurgau	Swiss Canton
TI	Ticino	Swiss Canton
UR	Uri	Swiss Canton
VD	Vaud	Swiss Canton
VS	Valais	Swiss Canton
ZG	Zug	Swiss Canton
ZH	Zurich	Swiss Canton

Appendix A

The following table lists the items used to construct the four attitudinal indices. Participants responded to these statements on a Likert scale. Items marked with (-) were reverse-coded for the analysis.

Construct/Variable	Original Item (German)	English Translation
Interactional adaptability (<i>aff_pers</i>)	Wenn ich mit Leuten aus einem anderen Dialektgebiet spreche, achte ich darauf, gewisse Wörter oder Wendungen zu vermeiden.	When speaking with people from another dialect area, I make sure to avoid certain words or phrases.
	Ich rede immer gleich, egal ob mein Gegenüber einen ähnlichen Dialekt spricht oder nicht. (-)	I always speak the same way, regardless of whether my interlocutor speaks a similar dialect or not. (-)
	Wenn ich mit fremden Leuten aus anderen Dialektregionen spreche, achte ich auf meine Wortwahl.	When speaking with strangers from other dialect regions, I pay attention to my word choice.
	Ich ertappe mich oft, wie ich Wörter und Wendungen brauche, die man in meinem Dialekt eigentlich nicht so sagt.	I often catch myself using words and phrases that are not actually said that way in my dialect.
	Wenn ich mit Leuten aus einem anderen Dialektgebiet spreche, passe ich mich an.	When speaking with people from another dialect area, I adapt.
Local attachment (<i>aff_identity</i>)	Ich bin stolz, ein/e Bewohner/in meines Wohnortes zu sein.	I am proud to be a resident of my municipality.
	Ich finde es schön, wenn Leute aus anderen Regionen merken, welchen Dialekt ich spreche.	I like it when people from other regions notice which dialect I speak.
	Ich finde es schade, wenn junge Leute viele Dialektwörter und -wendungen nicht mehr kennen.	I find it regrettable when young people no longer know many dialect words and phrases.
	Ich finde es schade, wenn ein Bekannter in eine andere Region zieht und nach wenigen Jahren schon nicht mehr unseren Dialekt spricht.	I find it regrettable when an acquaintance moves to another region and stops speaking our dialect after just a few years.
	Dialekte verändern sich halt, das stört mich nicht. (-)	Dialects simply change; that does not bother me. (-)
Standard orientation (<i>aff_sg</i>)	Hochdeutsch ist für mich eine Fremdsprache. (-)	Standard German is a foreign language to me. (-)
	Ich spreche gerne Hochdeutsch.	I like speaking Standard German.
	Wenn ich Hochdeutsch spreche, fühle ich mich wohl.	When I speak Standard German, I feel comfortable.
Metalinguistic awareness (<i>aff_change</i>)	Ich finde, schweizerdeutsche Dialekte haben sich in den letzten 50 Jahren stark verändert.	I think Swiss German dialects have changed significantly in the last 50 years.
	Ich befürchte, dass wir Deutschschweizer in 50 Jahren alle fast gleich sprechen.	I fear that in 50 years, all German-speaking Swiss will speak almost the same way.

Construct/Variable	Original Item (German)	English Translation
	Ich denke, schweizerdeutsche Dialekte werden sich in Zukunft stark verändern.	I think Swiss German dialects will change significantly in the future.
	Ich finde, junge Leute sprechen ganz anders als alte.	I think young people speak very differently from older people.

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